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RPOW

004 / 15 AUG 2004 / HAL FINNEY / RARE

WORK ONCE. PASS IT ON.

In a glowing workshop, one mouse strains to move a numbered puzzle tile while another sits above with a token in hand. The effort below and the reusable token above turn an abstract idea into an illustrated scene: work produces a token whose value can be passed on through exchange.

On 15 August 2004, Finney invited the cypherpunks mailing list to test his new server at rpow.net. It accepted a Hashcash proof-of-work token and returned an RSA-signed token he called a **Reusable Proof of Work**, or RPOW. A recipient could later exchange that RPOW for a fresh one of equal value. Each individual token could be spent only once, yet the chain of exchanges let the value move from person to person. As Finney put it, unlike Hashcash, **“they are reusable.”**

The promise depended on trust in the machine doing the exchange. Finney ran the server on an IBM 4758 secure cryptographic coprocessor. Its signed attestation was designed to let a sufficiently motivated user check the software configuration against the published source code. That was the safeguard behind his claim that the server would issue a new token only when it received an equal-value proof. RPOW was an experiment with a trusted server, not Bitcoin or a decentralised blockchain.

Finney called the system an early beta and warned users that a server reload might invalidate tokens already made. The mice, numbered puzzle, physical tokens and exchange machine in this illustration are fictional metaphors, not a reconstruction of the real RPOW hardware or a depiction of Finney himself.

DETAILS IN THE ART

01 THE SLIDING PUZZLE

The numbered tiles and empty space turn computational effort into a visible physical task. The puzzle is an artistic metaphor for work, not the actual computation performed by RPOW.

02 THE TOKEN IN HAND

The seated mouse holds a gold token while the other mouse works below. It represents value that can be passed on through exchange; the physical coin is fictional, not a documented RPOW object.

03 THE RECYCLING EMBLEM

The green recycling symbol on the dispenser points to reuse. In the real system, an individual token could be spent only once and exchanged for a fresh token of equal value; the machine here is a visual metaphor.

04 THE TWO MICE

The orange mouse with a backpack strains at the puzzle while the lighter mouse in a dark hoodie sits above. These fictional HISTROVE characters contrast effort and the resulting token, and are intended to return as small Easter eggs in future illustrations. They are not historical participants.

SOURCE & ART NOTE

Hal Finney, RPOW - Reusable Proofs of Work, 15 August 2004. The source describes an early beta on a secure server and warns that a reload could invalidate tokens. The mice, workshop, numbered puzzle, physical tokens and exchange apparatus are fictional illustrations. WORK ONCE. PASS IT ON. is the approved editorial caption, not a quotation from Finney.